TECHNICAL EVALUATION — SUBMERSIBLE PUMP

1. Evaluation output

Field	Value
Report	submersible_pump_EN201222034S.pdf (89 pages)
Standards	IEC 60335-1:2010 (Part 1); IEC 60335-2-41:2021 (Part 2 2-41)
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2. Report identity

Field	Value
Report number	EN201222034S
Laboratory	EMTEK(NINGBO) CO., LTD.
Report date	28 December 2025
Product declared	Submersible pump
Declared model	SPC55068
Standards claimed	IEC 60335-1:2010+A1:2013+A2:2016; IEC 60335-2-41:2025 on the test specification page; IEC 60335-2-41:2012 in the compliance statement
Standards evaluated	US IEC 60335-1:2010; EN IEC 60335-2-41:2021, incorporating IEC 60335-2-41:2012; supplied texts only

3. Summary

Field	Value
Final verdict	REPORT CORRECTION / REPLACEMENT REQUIRED — PRODUCT COMPLIANCE NOT DETERMINED — CERTIFICATION HOLD
Reason	The tested model and ratings conflict with the declared product, and several safety tests lack applicable or traceable evidence. The report’s standard references and page attribution are inconsistent.
Recommended action	Obtain a corrected or replacement report supported by identified-sample test records, any necessary retesting, and the applicable editions and amendments before certification proceeds.
Conclusion	The supplied report does not demonstrate compliance of model SPC55068 with the evaluated requirements. The recorded measurements cannot reliably be attributed to the declared 220–240 V, 550 W pump, and the missing evidence does not establish a product non-conformity; certification remains on hold pending resolution.
Findings	26 total — Critical 6, High 19, Medium 1, Low 0

4. Findings

Priority	Clause	Normative requirement	Finding reason	Required action	Pages	Evidence
Critical	4; 5.2	The appliance must withstand all relevant type tests.	The declared model is SPC55068, but the summary states that full tests were performed on 4TSC4.0-96-110/1000; no relationship between the models is established. The photograph on page 88 also displays 750 W, whereas the declared rating is 550 W.	Identify the tested sample and provide traceable test evidence for SPC55068, including its actual rating.	PDF pp.1–3,88	Report: verified p.3 Standard: verified in IEC 60335-1:2010 p.26
Critical	7.1	The appliance must be marked with its rated voltage or voltage range.	The declared and illustrated rating is 220–240 V, but the clause 7.1 result records 96 V; the report also describes the supply as DC while the marking specifies 50 Hz.	Establish the actual marked voltage and supply type and replace inconsistent records.	PDF pp.2–4,6	Report: verified p.6 Standard: verified in IEC 60335-1:2010 p.29
Critical	10.1	Marked rated power input is verified by measurement at rated voltage and normal operation.	The declared power is 550 W, but the power-input table uses a 1,000 W rated basis and records 995.2 W at 72 V. The table does not establish the power input of the declared model at its rated voltage.	Test the identified SPC55068 at its applicable rated voltage or voltages and report its measured power against the 550 W rating.	PDF pp.2–3,79	Report: verified p.79 Standard: verified in IEC 60335-1:2010 p.37
Critical	5.8.1; 11.5; 13.1; 16.2	Motor-operated appliances are tested at the specified voltages derived from their applicable rated voltage.	Heating and leakage tables record 76.3 V, rather than a voltage derived from the declared 220–240 V range; the supply type is also inconsistent. The recorded results cannot substantiate the declared rating.	Repeat or substantiate the tests on the identified model at the applicable supply type and required voltages.	PDF pp.2–4,79–80	Report: verified p.79 Standard: verified in IEC 60335-1:2010 p.45
Critical	—	The supplied European pump standard is read together with its identified A11 amendment.	The report cites IEC 60335-2-41:2025 in its specification but 2012 in its compliance statement; the claimed Part 1 amendments and the European A11 amendment were not supplied. Their requirements cannot be evaluated from the supplied texts.	Provide the claimed IEC editions and amendments and, if European conformity is sought, EN IEC 60335-2-41:2021/A11:2021, then reconcile the report's standard references.	PDF pp.2–3	Report: verified p.2 Standard: verified in IEC 60335-2-41:2021 p.4
Critical	—	Report pages and test results must be attributable to the report under review.	The footer on PDF page 86 carries report number EN200622010S rather than EN201222034S, undermining attribution of the creepage table and adjacent results. Several other page footers omit the total page count.	Obtain a laboratory-issued replacement with verified page attribution and complete numbering.	PDF pp.2,5 4,63,81,86	Report: verified p.86 Standard: requirement not in supplied standards
High	19.7	The locked-rotor test is conducted at rated voltage for the applicable duration, with winding temperature within the relevant limit.	The locked-rotor table records 116.6 V DC, inconsistent with the declared 220–240 V, 50 Hz rating. Test duration and protective-device operation are not recorded, so the selected 240 °C limit cannot be verified.	Repeat the locked-rotor test at the applicable rated supply and record duration, protection operation and applicable winding limit.	PDF pp.2,2 0–21,81	Report: verified p.81 Standard: verified in IEC 60335-1:2010 p.54
High	3.1.9; 10.1	Pumps are operated with zero inlet pressure and at the total head producing the highest power input.	The power-input table records "Max setting" but does not identify inlet pressure, total head or how the highest input was established.	Record the inlet pressure, total head and method used to establish the highest power input.	PDF pp.13,79	Report: verified p.79 Standard: verified in IEC 60335-2-41:2021 p.12
High	5.7; 11.7	Liquid temperature must be controlled relative to the marked temperature, and the heating test uses the marked liquid temperature.	The report claims P but gives no measured liquid temperature or definite marked maximum temperature; "<45□" does not establish the marked value or test condition.	Provide the actual temperature marking and recorded liquid temperatures for the relevant tests.	PDF pp.3,6 ,14,79	Report: verified p.3 Standard: verified in IEC 60335-2-41:2021 p.15
High	5.101; 25.14	Pumps are tested as portable unless fixed, and portable pumps other than the specified exceptions undergo the cord flexing test.	Both portable and fixed classifications are unchecked on the particulars page, while 5.101 and the 25.14 flexing test are N/A without a documented basis.	Establish whether the pump is fixed and perform the cord flexing test if it is portable.	PDF pp.4,6,43	Report: verified p.43 Standard: verified in IEC 60335-2-41:2021 p.13

Priority	Clause	Normative requirement	Finding reason	Required action	Pages	Evidence
High	7.1	Submersible pumps above 50 W must be marked with maximum operating depth of at least 1 m.	The declared 550 W submersible pump has maximum operating depth recorded as N/A, and the corresponding marking requirement is also N/A.	Establish and mark the maximum operating depth and record its verification.	PDF pp.2,4,7	Report: verified p.4 Standard: verified in IEC 60335-2-41:2021 p.14
High	7.1	Pumps must be marked with maximum liquid temperature of at least 35 °C.	The report awards P and refers to the marking plate, but its reproduced plate contains no maximum liquid temperature; the stated "<45□" is not a definite marked value.	Provide a legible actual marking showing the maximum liquid temperature and verify it.	PDF pp.3,7	Report: verified p.7 Standard: verified in IEC 60335-2-41:2021 p.14
High	15.1.2	The submersible pump is immersed for 24 h under the prescribed solution, temperature and pressure conditions.	A pass is recorded without the actual operating depth, test pressure, water temperature or salt concentration. The checklist also refers to 5 m instead of the supplied standard's 15 m alternative; the actual pressure applied is not established.	Provide the complete immersion-test conditions and repeat the test at the required pressure if necessary.	PDF pp.4,17	Report: verified p.17 Standard: verified in IEC 60335-2-41:2021 p.16
High	19.101; 19.13	Pumps undergo the specified dry-running sequence, followed by the applicable abnormal-operation assessment.	The checklist records P for the sequence, but the appended abnormal-operation table does not identify the 19.101 test conditions, actual head, test voltage, observations or results.	Provide traceable 19.101 test records and associated 19.13 observations for the identified sample.	PDF pp.25, 81–82	Report: verified p.25 Standard: verified in IEC 60335-2-41:2021 p.17
High	24.1.3	Level switches are subjected to 50,000 operating cycles.	The level-switch test is N/A, but the product photograph shows an apparent float device and the report does not identify its function or justify the exclusion.	Identify the photographed device and provide the 50,000-cycle assessment if it is a level switch.	PDF pp.38,88	Report: verified p.38 Standard: verified in IEC 60335-2-41:2021 p.19
High	24.1.4	A thermal motor protector necessary for compliance is tested with its motor under Annex D.	The particulars page indicates a self-resetting thermal cut-out, while 24.1.4 and Annex D are N/A without identifying the device or establishing whether it is needed for compliance.	Identify the thermal cut-out, establish its safety function and provide the applicable component and motor-protector evidence.	PDF pp.4,38,62	Report: verified p.38 Standard: verified in IEC 60335-1:2010 p.77
High	25.1; 25.3	Submersible pumps other than class III must have a flexible supply cord fitted with a plug.	The pump is declared class I and submersible, but the specific 25.1 and 25.3 entries are N/A; the general plug entry is P while the particulars page does not indicate a fitted plug.	Verify and document the fitted flexible cord and plug on the identified sample.	PDF pp.4,40–41	Report: verified p.40 Standard: verified in IEC 60335-2-41:2021 p.19
High	25.7	Outdoor pumps other than class III require the specified polychloroprene or equivalent supply cord, subject to the stated exceptions.	Outdoor use is indicated on the particulars page, but the supplied report does not address the outdoor-cord requirement or identify the cord type.	Identify the supply cord and demonstrate compliance with the applicable outdoor-cord requirement.	PDF pp.4,41–42	Report: no quote given Standard: verified in IEC 60335-2-41:2021 p.19
High	25.8	A class I outdoor submersible pump requires a cord at least 10 m long or 3 m longer than marked operating depth, whichever is greater.	Outdoor use and class I submersible construction are indicated, but the cord-length entry is N/A and neither cord length nor operating depth is recorded.	Measure the cord and verify its length against the established operating depth.	PDF pp.4,42	Report: verified p.42 Standard: verified in IEC 60335-2-41:2021 p.19
High	25.8	Supply-cord conductors must meet the minimum nominal cross-sectional area in Table 11.	The conductor-size entry is N/A despite a supply cord being indicated; neither rated current nor conductor area is recorded for assessment.	Record the rated current and supply-cord conductor area and verify the applicable Table 11 minimum.	PDF pp.4,42	Report: verified p.42 Standard: verified in IEC 60335-1:2010 p.81
High	25.10; 27.3	A class I supply cord must have a connected green/yellow earthing core, arranged so current-carrying conductors become taut first if the cord slips.	Both requirements are N/A although the report identifies a class I appliance with a supply cord; no construction-based justification is given.	Inspect and document the protective-earthing core, connections and conductor arrangement.	PDF pp.4,43,49	Report: verified p.43 Standard: verified in IEC 60335-1:2010 p.90

Priority	Clause	Normative requirement	Finding reason	Required action	Pages	Evidence
High	27.5	Earthing continuity is tested at the specified current and calculated resistance must not exceed 0.1 Ω.	A pass is reported without the measured resistance, test current or identified metal parts.	Record the test current and measured resistance for each relevant earthed part.	PDF p.49	Report: verified p.49 Standard: verified in IEC 60335-1:2010 p.91
High	29.1; 29.2	Clearances and creepage distances must be assessed for the relevant voltage, insulation, pollution degree and material group.	The tables use a 500 V impulse-voltage row and a ≤50 V working-voltage row despite the declared 220–240 V rating; the assessed locations and material group are not identified, and functional creepage measurements are absent.	Reassess and document each relevant insulation path using the verified product rating, working voltage and material group.	PDF pp.2,53–54,83,85	Report: verified p.85 Standard: verified in IEC 60335-1:2010 p.99
High	30.1	Relevant non-metallic and insulating parts whose deterioration could affect safety must be sufficiently heat resistant.	Clause 30.1 is N/A without an applicability assessment, although an insulating bushing is listed among critical components. The heat-test table contains no results.	Identify relevant parts and provide the applicable heat-resistance assessment or a substantiated exclusion.	PDF pp.55, 82,87	Report: verified p.55 Standard: verified in IEC 60335-1:2010 p.102
High	30.2; 30.2.1; 30.2.3	Applicable non-metallic parts must be assessed for ignition and fire spread using the pump-specific test route.	All clause 30.2 routes are N/A, and the appended table is blank; the report does not establish the metal-enclosure and RCD-instruction conditions for the reduced submersible-pump route or otherwise justify excluding relevant parts.	Provide the applicable fire-resistance assessment, including evidence for any permitted test-route exemption.	PDF pp.56–58,87	Report: verified p.56 Standard: verified in IEC 60335-2-41:2021 p.20
Medium	7.12.1	Installation instructions must state maximum total head for pumps rated above 50 W.	A pass refers to marking detail, but neither a maximum head value nor the installation instructions are supplied in the report.	Provide the installation instructions and identify the maximum total head in metres.	PDF pp.3,10	Report: verified p.10 Standard: verified in IEC 60335-2-41:2021 p.14